

FILTERING IN THE FREQUENCY DOMAIN

1. 2-D DFT SEPARABILITY

The 2-D DFT is separable.

It can be computed using successive 1-D DFTs.

Process:

1. Compute N-point DFT along each row: $G(m,v) = \sum [X(m,n) * e^{-j2\pi(vn/N)}]$
2. Compute M-point DFT along each column of $G(m,v)$: $F(u,v) = \sum [G(m,v) * e^{-j2\pi(um/M)}]$
3. KEY DFT PROPERTIES

Translation (Frequency Shift):

Multiplying $s(n)$ by a complex exponential $e^{j(2\pi k_1 n/N)}$ in the time domain shifts its DFT in the frequency domain.

$$h(n) = s(n) * e^{j(2\pi k_1 n/N)} \rightarrow H(k) = S(k - k_1)$$

Translation (Spatial Shift):

$$f(x - x_0, y - y_0) \leftrightarrow F(u,v) * e^{-j2\pi((ux_0/M) + (vy_0/N))}$$

Convolution Theorem:

Convolution in spatial domain $\leftrightarrow$ Multiplication in frequency domain.

$$f(x,y) * h(x,y) \leftrightarrow F(u,v) \cdot H(u,v)$$

This multiplication is element-wise (pointwise), not standard matrix multiplication.

Delta Function Symmetry:

$$\delta(c-x) = \delta(x-c)$$

Cosine/Delta Relationship:

$$\cos(2\pi u/M) \leftrightarrow 0.5 * [\delta(x+1,y) + \delta(x-1,y)]$$

$$\cos(2\pi v/N) \leftrightarrow 0.5 * [\delta(x,y+1) + \delta(x,y-1)]$$

3. LAPLACIAN OPERATOR

Spatial Domain: $\nabla^2 f(x,y) = (\partial^2 f / \partial x^2) + (\partial^2 f / \partial y^2)$. It measures second-order variations.

Frequency Domain: $F\{\nabla^2 f\} = -4\pi^2(u^2 + v^2)F(u,v)$.

Laplacian Filter Formulas:

Discrete, Zero-centered (0,0): $H(u,v) = -(u^2 + v^2)$

Discrete, Centered (M/2, N/2): $H(u,v) = -[(u - M/2)^2 + (v - N/2)^2]$

4. FILTER CONVERSION EXAMPLE (FREQ $\rightarrow$ SPATIAL)

Given Filter: $H(u,v) = 0.8 - 0.3\cos(2\pi u/M) - 0.5\cos(2\pi v/N)$

Method:

1. Use linearity to split terms.
2. Apply inverse DFT to cosine terms (results in delta functions).
3. Apply inverse DFT to constant (results in delta at origin): $C \leftrightarrow C * \delta(x,y)$.

Resulting Spatial Filter: $h(x,y) = 0.8\delta(x,y) - 0.15[\delta(x+1,y) + \delta(x-1,y)] - 0.25*[\delta(x,y+1) + \delta(x,y-1)]$

This creates a 3x3 mask.

5. HOMOMORPHIC FILTERING

Purpose: Reduce uneven illumination, enhance details.

Image Model: $f(x,y) = i(x,y) \cdot r(x,y)$

$i(x,y)$: Illumination (low frequency)

$r(x,y)$: Reflectance (high frequency)

Process:

1. Log Transform: Converts multiplication to addition. $\ln[f(x,y)] = \ln[i(x,y)] + \ln[r(x,y)]$.
2. Fourier Transform: $F(u,v) = F\{\ln[f(x,y)]\} = I(u,v) + R(u,v)$.
3. High-Pass Filtering: Apply filter $H(u,v)$ (e.g., Butterworth, Gaussian) to suppress $I(u,v)$ and enhance $R(u,v)$. $G(u,v) = H(u,v) \cdot F(u,v)$.
4. Inverse Fourier Transform: $g(x,y) = F^{-1}\{G(u,v)\}$. (Result is still in log domain).
5. Exponential Transform: Undoes the log. $f_{\text{enhanced}}(x,y) = \exp[g(x,y)]$.

6. FILTER FORMULAS

Butterworth:

Low-Pass (LP): $H_{\text{LP}}(u,v) = 1 / (1 + (D(u,v)/D_0)^{2n})$

High-Pass (HP): $H_{\text{HP}}(u,v) = 1 / (1 + (D_0/D(u,v))^{2n})$

Relationship: $H_{\text{LP}}(u,v) = 1 - H_{\text{HP}}(u,v)$ (They are complementary).

Gaussian:

Low-Pass (GLPF): $H_{\text{LP}}(u,v) = e^{(-D^2(u,v) / 2D_0^2)}$

High-Pass (GHPF): $H_{\text{HP}}(u,v) = 1 - e^{(-D^2(u,v) / 2D_0^2)}$